Ira A. Fulton Schools of Engineering
Arizona State University

CSE 579: Knowledge Representation and Reasoning

Automated Warehouse Scenario Project Part 1

Purpose

The course project will have four (4) requirements. The aim of part 1 is for you to become familiar with clingo, a powerful tool used to run answer set programs. After completing parts 1 and 2 you will be able to run logic programs for this project.

Objectives

Learners will be able to:

- Set up a Clingo environment.
- Understand basic functionality and use of Clingo for answer set programs.

Technology Requirements

- Clingo Download Tutorial - Github
- Clingo Online

Project Description

This is the first milestone for the course projects. Each project will use Clingo. This assignment allows learners to install and begin using Clingo.

Directions

While you may run Clingo online you must install Clingo on your local machine to complete the graded assignments in Weeks 4 and 5 and the course project. Potassco, the Posdam Answer Set Solving Collection, provides extensive documentation, including a Getting Started Guide
There is also a PDF resource to for more information on Installing and Trying out Clingo.

To facilitate your initiation with clingo, kindly adhere to the three steps outlined in the quiz, utilizing the provided prompts.

Step 1:

1. Consult the clingo installation guide in the quiz to download clingo onto your personal machine. Execute a test on the example program provided. Note that the example clingo program:

```
1{p(1..6)}.
```

is referred to as a choice rule with a cardinality constraint. While the intricacies of this concept will be covered in Module 4, a comprehensive understanding is not imperative for the completion of Project Milestone 1. This program has been intentionally selected so that you can exclusively determine the number of stable models through clingo.

Upon completing Step 1, generate a "test.txt" file, which will be utilized in the subsequent step of this milestone.

Step 2:

A. In the preceding step, you saved the example program in a file named "test.txt" and employed clingo to identify all its stable models.
B. In this step, it is essential to compare your input program (the clingo program stored in "test.txt"), command line, and output with the corresponding entries in the provided table. Any disparities may indicate potential issues, necessitating a thorough review.

NOTE: The command line format may vary based on your operating system and file path. In future assignments, you may truncate lengthy outputs, but ensure that the pasted output is substantial and supports your answers in the assignment to the best extent possible.

InputProgram	1{p(1..6)}.
CommandLine	clingo test.txt 0

Output(truncated)	Solving... Answer: 1 p(6)... Answer: 63 p(1) p(2) p(3) p(4) p(5) p(6) SATISFIABLE Models : 63

Submission Directions for Project Deliverables

You are given a limited attempts to submit your best work. The number of attempts is given to anticipate any submission errors you may have in regards to properly submitting your best work within the deadline (e.g., accidentally submitting the wrong paper). It is not meant for you to receive multiple rounds of feedback and then one (1) final submission. Only your most recent submission will be assessed.

You must submit your Automated Warehouse Scenario Project Part 1 deliverable through Canvas. Carefully review submission directions outlined in this overview document in order to correctly earn credit for your work. Learners may not email or use other means to submit any assignment or project for review, including feedback, and grading.

The Automated Warehouse Scenario Project Part 1 Project includes one (1) deliverables:

- **Quiz:** Submit the Automated Warehouse Scenario Project Part 1 quiz.

You will be assessed on your completion and answers to the questions in the quiz. There will be one multiple-choice, application question in this milestone.

You will have 180 minutes to complete this milestone and three (3) attempts to submit your work.To submit your work follow the directions below:

1. Navigate to your course and click on the "**Submission: Automated Warehouse Scenario Project Part 1**" submission space
2. Click "**Take the quiz**" in the upper right corner
3. Complete each question.
4. When you are finished, click "**Submit Quiz.**"
    a. If you would like to retake the quiz, click "**Take the Quiz Again.**"

Evaluation

The only evaluation criterion for this deliverable is the completion of the question on the Automated Warehouse Scenario Project Part 1 submission space, and this milestone is worth fifteen (15) points.